

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk:

Product Description:

Cat No. :

**Kloroform**
**Chloroform**

C/4960/25, C/4960/27, C/4960/17, C/4960/15, C/4960/08, C/4960/PB08, C/4960/PB15, C/4960/PB17, C/4960/21RSS, C/4960/24RSS, C/4960/25RSS, C/4960/34RSS, C/4960/27RSS, C/4960/15PC

Synonyms

Methane trichloride; Methenyl trichloride; Formyl trichloride

CAS No

67-66-3

Molecular Formula

 C H Cl<sub>3</sub>
**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use

Laboratory chemicals.

Uses advised against

**Company**

 Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**Supplier**

E-mail address

Enquiry.my@thermofisher.com

Emergency Telephone Number

 Tel: +03-5525 7888  
 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                      |                    |
|------------------------------------------------------|--------------------|
| Acute oral toxicity                                  | Category 4 (H302)  |
| Acute Inhalation Toxicity - Vapors                   | Category 3 (H331)  |
| Skin Corrosion/Irritation                            | Category 2 (H315)  |
| Serious Eye Damage/Eye Irritation                    | Category 2 (H319)  |
| Carcinogenicity                                      | Category 2 (H351)  |
| Reproductive Toxicity                                | Category 2 (H361d) |
| Specific target organ toxicity - (single exposure)   | Category 3 (H336)  |
| Specific target organ toxicity - (repeated exposure) | Category 1 (H372)  |

**Label Elements**

# SAFETY DATA SHEET

Chloroform

Revision Date 23-Mar-2025



**Signal Word**

**Danger**

## Hazard Statements

H302 - Harmful if swallowed  
H331 - Toxic if inhaled  
H315 - Causes skin irritation  
H319 - Causes serious eye irritation  
H336 - May cause drowsiness or dizziness  
H351 - Suspected of causing cancer  
H361d - Suspected of damaging the unborn child  
H372 - Causes damage to organs through prolonged or repeated exposure

## Precautionary Statements

### Prevention

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear eye protection/ face protection

### Response

P301 + P312 - IF SWALLOWED: Call a POISON CENTER or doctor if you feel unwell  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P308 + P313 - IF exposed or concerned: Get medical advice/attention  
P330 - Rinse mouth  
P332 + P313 - If skin irritation occurs: Get medical advice/attention  
P362 + P364 - Take off contaminated clothing and wash it before reuse

### Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P405 - Store locked up

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

Cardiac and respiratory depression  
Overexposure may cause decreased heart rate, decreased blood pressure, heart block, and cardiac failure  
Toxic to terrestrial vertebrates  
This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component  | CAS No   | Weight % |
|------------|----------|----------|
| Chloroform | 67-66-3  | >99      |
| 1-Pentene  | 109-67-1 | 0.01     |

# SAFETY DATA SHEET

Chloroform

Revision Date 23-Mar-2025

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### **General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### **Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

#### **Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

#### **Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

#### **Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

#### **Self-Protection of the First Aider**

Use personal protective equipment as required.

### Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Symptoms of overexposure are dizziness, headache, tiredness, nausea, unconsciousness, cessation of breathing. Causes central nervous system depression.

### Indication of any immediate medical attention and special treatment needed

#### **Notes to Physician**

Treat symptomatically. Signs of overdose include stupor and respiratory depression. Symptoms may be delayed.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Substance is nonflammable; use agent most appropriate to extinguish surrounding fire.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

Non-combustible, substance itself does not burn but may decompose upon heating to produce corrosive and/or toxic fumes.

#### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Phosgene, Hydrogen chloride gas.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

# SAFETY DATA SHEET

Chloroform

Revision Date 23-Mar-2025

## Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

## Environmental precautions

Should not be released into the environment.

## Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

## Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Protect from direct sunlight. Store under an inert atmosphere. Protect from moisture.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component  | Malaysia | ACGIH TLV   | OSHA PEL                                                                                                           |
|------------|----------|-------------|--------------------------------------------------------------------------------------------------------------------|
| Chloroform |          | TWA: 10 ppm | (Vacated) TWA: 2 ppm<br>(Vacated) TWA: 9.78 mg/m <sup>3</sup><br>Ceiling: 50 ppm<br>Ceiling: 240 mg/m <sup>3</sup> |

| Component  | European Union                                                                                          | The United Kingdom                                                                      | Germany                                          |
|------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------|
| Chloroform | TWA: 2 ppm 8 hr<br>TWA: 10 mg/m <sup>3</sup> 8 hr<br>Possibility of significant uptake through the skin | TWA: 2 ppm<br>TWA: 9.9 mg/m <sup>3</sup><br>STEL: 6 ppm<br>STEL: 29.7 mg/m <sup>3</sup> | 0.5 ppm TWA MAK<br>2.5 mg/m <sup>3</sup> TWA MAK |

### Exposure Controls

#### Engineering Measures

Use only under a chemical fume hood. Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

# SAFETY DATA SHEET

Chloroform

Revision Date 23-Mar-2025

## Personal protective equipment

|                                 |                       |
|---------------------------------|-----------------------|
| <b>Eye Protection</b>           | Goggles               |
| <b>Hand Protection</b>          | Protective gloves     |
| <b>Skin and body protection</b> | Long sleeved clothing |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                 |                                                                                                                                                                                                                                                    |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Respiratory Protection</b>   | When workers are facing concentrations above the exposure limit they must use appropriate certified respirators                                                                                                                                    |
| <b>Recommended Filter type:</b> | low boiling organic solvent Type AX Brown conforming to EN371<br>To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly<br>When RPE is used a face piece Fit Test should be conducted |

|                                |                                                                       |
|--------------------------------|-----------------------------------------------------------------------|
| <b><u>Hygiene Measures</u></b> | Handle in accordance with good industrial hygiene and safety practice |
|--------------------------------|-----------------------------------------------------------------------|

|                                               |                                      |
|-----------------------------------------------|--------------------------------------|
| <b><u>Environmental exposure controls</u></b> | Prevent product from entering drains |
|-----------------------------------------------|--------------------------------------|

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>                              | Colorless                |                                          |
| <b>Physical State</b>                          | Liquid                   |                                          |
| <b>Odor</b>                                    | aromatic sweet           |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>pH</b>                                      | No information available |                                          |
| <b>Melting Point/Range</b>                     | -63 °C / -81.4 °F        |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | 61 °C / 141.8 °F         |                                          |
| <b>Flash Point</b>                             | No information available | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | No data available        |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Vapor Pressure</b>                          | 213 mbar @ 20 °C         |                                          |
| <b>Vapor Density</b>                           | No data available        | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>              | 1.480                    |                                          |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                   |
| <b>Water Solubility</b>                        | 8 g/L (20°C)             |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Component</b>                               | <b>log Pow</b>           |                                          |
| Chloroform                                     | 2                        |                                          |

# SAFETY DATA SHEET

Chloroform

Revision Date 23-Mar-2025

1-Pentene 2.66

Autoignition Temperature No data available  
Decomposition Temperature No data available  
Viscosity 0.56 mPa s at 20 °C  
Explosive Properties No information available  
Oxidizing Properties No information available

Molecular Formula C H Cl<sub>3</sub>  
Molecular Weight 119.38  
VOC Content(%) 100

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions. UNSTABLE (REACTIVE) UPON DEPLETION OF INHIBITOR. Light sensitive.

### Possibility of Hazardous Reactions

Hazardous Polymerization Hazardous polymerization does not occur.  
Hazardous Reactions None under normal processing.

### Conditions to Avoid

Incompatible products. Heat, flames and sparks. Excess heat. Exposure to light. Protect from moisture.

### Incompatible Materials

Strong oxidizing agents. Alkali metals. Aluminium. Acetone.

### Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Phosgene. Hydrogen chloride gas.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

(a) acute toxicity;  
Oral Category 4  
Dermal Based on available data, the classification criteria are not met  
Inhalation Category 3

# SAFETY DATA SHEET

Chloroform

Revision Date 23-Mar-2025

| Component  | LD50 Oral                                                                      | LD50 Dermal               | LC50 Inhalation              |
|------------|--------------------------------------------------------------------------------|---------------------------|------------------------------|
| Chloroform | LD50 = 908 mg/kg (rat)<br>LD50 = 695 mg/kg ( Rat )<br>LD50 = 450 mg/kg ( Rat ) | LD50 > 20 g/kg ( Rabbit ) | LC50 = 10.5 mg/L ( Rat ) 4 h |
| 1-Pentene  | >2000 mg/kg (Rat)                                                              | >2000 mg/kg (Rabbit)      | LC50 = 10000 ppm ( Rat ) 4 h |

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; Category 2

(d) respiratory or skin sensitization;

Respiratory  
Skin

Based on available data, the classification criteria are not met  
Based on available data, the classification criteria are not met

(e) germ cell mutagenicity;

Based on available data, the classification criteria are not met

(f) carcinogenicity;

Category 2

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component  | EU | UK | Germany | IARC     |
|------------|----|----|---------|----------|
| Chloroform |    |    |         | Group 2B |

(g) reproductive toxicity;

Reproductive Effects  
Developmental Effects  
Teratogenicity

Category 2

Experiments have shown reproductive toxicity effects on laboratory animals.  
Developmental effects have occurred in experimental animals.  
Study result . negative.

(h) STOT-single exposure;

Category 3

Results / Target organs

Central nervous system (CNS).

(i) STOT-repeated exposure;

Category 1

Study result

LOAEL = 15 mg/kg bw/day  
NOAEC = 25 mg/m<sup>3</sup>

Route of exposure  
Target Organs

Inhalation  
Liver, Kidney.

(j) aspiration hazard;

Based on available data, the classification criteria are not met

Other Adverse Effects

Tumorigenic effects have been reported in experimental animals.

Symptoms / effects, both acute and delayed

Symptoms of overexposure are dizziness, headache, tiredness, nausea, unconsciousness, cessation of breathing. Causes central nervous system depression.

Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects

Do not empty into drains. The product contains following substances which are hazardous for the environment. Contains a substance which is: Harmful to aquatic organisms.

# SAFETY DATA SHEET

Chloroform

Revision Date 23-Mar-2025

| Component  | Freshwater Fish                                                                                                                                                                                                                     | Water Flea           | Freshwater Algae    | Microtox                                                                                                                                                     |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chloroform | LC50: = 300 mg/L, 96h static (Poecilia reticulata)<br>LC50: = 18 mg/L, 96h flow-through (Lepomis macrochirus)<br>LC50: = 18 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: = 71 mg/L, 96h flow-through (Pimephales promelas) | EC50 = 28.9 mg/L/48h | EC50 = 560 mg/L/48h | Photobacterium phosphoreum: EC50 = 520 mg/L/5 min<br>Photobacterium phosphoreum: EC50 = 670 mg/L/15 min<br>Photobacterium phosphoreum: EC50 = 670 mg/L/30min |

## Persistence and degradability

### **Persistence**

### **Degradation in sewage treatment plant**

Product is biodegradable

Persistence is unlikely, based on information available.

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

## Bioaccumulative potential

Bioaccumulation is unlikely

| Component  | log Pow | Bioconcentration factor (BCF) |
|------------|---------|-------------------------------|
| Chloroform | 2       | 1.4 - 13 dimensionless        |
| 1-Pentene  | 2.66    | No data available             |

## Mobility in soil

The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air.

## Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

## Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

### **Waste from Residues/Unused Products**

Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

### **Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

### **Other Information**

Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No

UN1888

Hazard Class

6.1

Packing Group

III

Proper Shipping Name

Chloroform



# SAFETY DATA SHEET

Chloroform

Revision Date 23-Mar-2025

## Road and Rail Transport

UN-No UN1888  
Hazard Class 6.1  
Packing Group III  
Proper Shipping Name Chloroform

## IATA

UN-No UN1888  
Hazard Class 6.1  
Packing Group III  
Proper Shipping Name Chloroform

Special Precautions for User No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories X = listed

| Component  | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Chloroform | 200-663-8 | X    | X   | X     | X    | X    | X     | X    | X        |
| 1-Pentene  | 203-694-5 | X    | X   | X     | X    | X    | X     | X    | KE-28027 |

**Note** Amylene is used as a stabilizer, but there is evidence that it may not prevent phosgene generation. Chloroform stabilized with amylene should be tested for phosgene content.

| Component  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Chloroform |                                                                                           |                                                                                          |                            | Annex I - Y45                      |

### National Regulations

**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

FSUC4960

# SAFETY DATA SHEET

Chloroform

Revision Date 23-Mar-2025

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Revision Date

23-Mar-2025

Revision Summary

SDS sections updated, 7.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**